

Architecture Type	Instruction Length	Complexity Type	Memory-Register	Microarchitecture	Performance Imp	Target Application	Source
CISC (x86 / Intel)	Variable (1 to 15)	Complex (internal)	Complex (support)	Micro-op decoding	High performance	Personal computer	1-4
Intel x86 (Skylake)	Variable-length (CISC)	Internal CISC-to-RISC	Not in source	Micro-op cache (L1)	Hit rates up to 80%	General-purpose	5, 6
AMD Zen	Variable-length (CISC)	CISC-to-RISC transition	Not in source	Competitively shared	Improved execution	General-purpose	5
x86 (CISC Super)	Variable	Large set of complex	Complex addressing	Macro-op fusion, micro-op	High throughput	General-purpose	7
CRISC (Hybrid)	Variable (external)	Blend of CISC and RISC	Clean memory model	Deep pipelining, micro-op	Narrowed the gap	Modern mobile applications	8
Hybrid (NexGen)	Variable (external)	CISC (external) / RISC (internal)	Load/Store (internal)	Asynchronous x86	Efficiency of RISC	General purpose	4, 9-11
RISC (ARM / AA)	Fixed (32-bit) / Simple	Reduced (with complex)	Load/Store	Pipelining, hardware	High power efficiency	Mobile devices, servers	1, 12, 13
ARM (Apple Silicon)	Fixed (ARM standard)	Heterogeneous	Unified Memory Architecture	NEON vector unit	High power efficiency	Consumer laptops	14
RISC-V (General)	4-byte (Fixed) / Simple	Simple standard	Load/Store (via fixed)	Macro-op fusion	RV64GC fetches	Microcontrollers	15
RISC-V (Esperanto)	64-bit RISC-V internal	General-purpose	Distributed on-chip	Low-voltage operation	Over 100 times better	Machine Learning	16
RISC (X-HEEP)	Variable (Support)	Reduced / Open	Load/Store	eXtensible Accelerator	Up to 7.3× speed	TinyAI, ultra-low-power	17
RISC (RISC-V G)	Fixed (32-bit based)	Reduced with custom	Load/Store	Custom HDC instructions	Up to 56.2× speed	HDC-CNN Hybrid	18
RISC	Fixed-length	Reduced/Simplified	Load/Store (Register)	Instruction pipelining	Predictable instructions	Embedded systems	8, 19-21
CISC	Variable-length	Complex instructions	Register-Memory	Microcode-based	High code density	Desktop PCs, computers	8, 19-21
VLIW (Transmeta)	64-bit, 128-bit micro	Code Morphing	Load/Store (64 general)	Aggressive speculation	60–70% power reduction	Low-power laptops	22-25
Hybrid PUM (DA)	ISA instructions (CISC)	General-purpose	Mixed (ACE row)	Hardware instructions	59.4x speedup for	Cryptography (AI)	26
Analog PUM (DA)	Not in source	Domain-specific	In-memory electronic	Bit-slicing, parallel	Fast but approximate	Machine learning	26
Digital PUM (DA)	Not in source	Boolean complete	In-memory Boolean	Bit-pipelining (bit)	Error-resilient; high	Data-intensive workloads	26
microx86 (RISC-V)	Variable (x86 encoding)	Reduced set of complex	Load-Store	Excluded instructions	Up to 9.8% peak	Power-efficient devices	7
Tensix (Tenstorrent)	Minimal RISC instructions	Specialised AI accelerators	Local SRAM (L1)	Circular buffers, zero-wait	Optimised for deep learning	Artificial Intelligence	27
Arm (Cortex-A72)	Not in source	High-performance	Not in source	Out-of-order execution	High performance	Processing systems	28
Arm (Cortex-M series)	Not in source	Low-power cores	Not in source	Not in source	Limited performance	Heterogeneous systems	28